Appendix 1

Searching strategy was shown:

PubMed

((((((((((((("Estrogens"[Mesh]) OR "Estrogens, Conjugated (USP)"[Mesh]) OR "Estradiol"[Mesh]) OR "estradiol valerate, cyproterone acetate drug combination" [Supplementary Concept]) OR "Progestins"[Mesh]) OR "Hormones"[Mesh]) OR "Hormones, Hormone Substitutes, and Hormone Antagonists"[Mesh]) OR "Gonadal Hormones"[Mesh]) OR "Gonadal Steroid Hormones"[Mesh]) OR "Estrogen Receptor Modulators"[Mesh]) OR "Estrogen Replacement Therapy"[Mesh]) OR "Raloxifene Hydrochloride"[Mesh]) OR "Hormone Replacement Therapy"[Mesh]

(((((("Carcinoma, Bronchogenic"[Mesh]) OR ("Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] OR "Lung Neoplasms"[Mesh]))

(((((("Carcinoma, Bronchogenic"[Mesh]) OR ("Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] OR "Lung Neoplasms"[Mesh])) OR ((((((((((((((((((((((((((((((((Carcinoma, Non Small Cell Lung[Title/Abstract]) or(Carcinomas, Non-Small-Cell Lung[Title/Abstract])) OR (Carcinomas, Non-Small-Cell Lung[Title/Abstract])) OR (Lung Carcinomas, Non-Small-Cell[Title/Abstract])) OR (Non-Small-Cell Lung Carcinomas[Title/Abstract])) OR (Nonsmall Cell Lung Cancer[Title/Abstract])) OR (Non-Small-Cell Lung Carcinoma[Title/Abstract])) OR (Non Small Cell Lung Carcinoma[Title/Abstract])) OR (Carcinoma, Non-Small Cell Lung[Title/Abstract])) OR (Non-Small Cell Lung Cancer[Title/Abstract])) OR (Pulmonary Neoplasms[Title/Abstract])) OR (Neoplasms, Lung[Title/Abstract])) OR (Lung Neoplasm[Title/Abstract])) OR (Neoplasm, Lung[Title/Abstract])) OR (Neoplasms, Pulmonary[Title/Abstract])) OR (Neoplasm, Pulmonary[Title/Abstract])) OR (Pulmonary Neoplasm[Title/Abstract])) OR (Lung Cancer[Title/Abstract])) OR (Cancer, Lung[Title/Abstract])) OR (Cancers, Lung[Title/Abstract])) OR (Lung Cancers[Title/Abstract])) OR (Pulmonary Cancer[Title/Abstract])) OR (Cancer, Pulmonary[Title/Abstract])) OR (Cancers, Pulmonary[Title/Abstract])) OR (Pulmonary Cancers[Title/Abstract])) OR (Cancer of the Lung[Title/Abstract])) OR (Cancer of Lung[Title/Abstract])) OR (Small Cell Lung Cancer[Title/Abstract])) OR (Oat Cell Lung Cancer[Title/Abstract])) OR (Small Cell Cancer Of The Lung[Title/Abstract])) OR (Carcinoma, Small Cell Lung[Title/Abstract])) OR (Oat Cell Carcinoma of Lung[Title/Abstract])) OR (Carcinoma, Bronchogenic[Title/Abstract])))))) AND ((("estrogenic compounds"[Title/Abstract] OR "compounds estrogenic"[Title/Abstract] OR "estrogenic agents"[Title/Abstract] OR "agents estrogenic"[Title/Abstract] OR "Estrogen"[Title/Abstract] OR "estrogen receptor agonists"[Title/Abstract] OR "agonists estrogen receptor"[Title/Abstract] OR "agonists estrogen"[Title/Abstract] OR "estrogen effect"[Title/Abstract] OR "estrogenic effect"[Title/Abstract] OR "estrogenic effects"[Title/Abstract] OR "effects estrogenic"[Title/Abstract] OR "estrogen effects"[Title/Abstract] OR "effects estrogen"[Title/Abstract]) OR ((((((((((((((((((((((((Estradiol[Title/Abstract])) OR (Hormones[Title/Abstract])) OR (Hormones, Hormone Substitutes,[Title/Abstract] OR Hormone Antagonists[Title/Abstract])) OR (Progestins[Title/Abstract])) OR (Therapy, Hormone Replacement[Title/Abstract])) OR (Hormone Replacement Therapies[Title/Abstract])) OR (Replacement Therapies, Hormone[Title/Abstract])) OR (Therapies, Hormone Replacement[Title/Abstract])) OR (Replacement Therapy, Hormone[Title/Abstract])) OR (Estrogen Replacement Therapies[Title/Abstract])) OR (Replacement Therapies, Estrogen[Title/Abstract])) OR (Therapies, Estrogen Replacement[Title/Abstract])) OR (Therapy, Estrogen Replacement[Title/Abstract])) OR (Replacement Therapy, Estrogen[Title/Abstract])) OR (Estrogen Replacement[Title/Abstract])) OR (Estrogen Replacements[Title/Abstract])) OR (Replacements, Estrogen[Title/Abstract])) OR (Replacement, Estrogen[Title/Abstract])) OR (Postmenopausal Hormone Replacement Therapy[Title/Abstract])) OR (Hormone Replacement Therapy, Post-Menopausal[Title/Abstract])) OR (Hormone Replacement Therapy, Post

Menopausal[Title/Abstract])) OR (Estrogen Progestin Replacement Therapy[Title/Abstract])) OR (Estrogen Progestin Combination Therapy[Title/Abstract]))) OR ((((((((((((("Estrogens"[Mesh]) OR "Estrogens, Conjugated (USP)"[Mesh]) OR "Estradiol"[Mesh]) OR "estradiol valerate, cyproterone acetate drug combination" [Supplementary Concept]) OR "Progestins"[Mesh]) OR "Hormones"[Mesh]) OR "Hormones, Hormone Substitutes, and Hormone Antagonists"[Mesh]) OR "Gonadal Hormones"[Mesh]) OR "Gonadal Steroid Hormones"[Mesh]) OR "Estrogen Receptor Modulators"[Mesh]) OR "Estrogen Replacement Therapy"[Mesh]) OR "Raloxifene Hydrochloride"[Mesh]) OR "Hormone Replacement Therapy"[Mesh]))

Web of Science

(TS=(Estrogenic Compounds OR Compounds, Estrogenic OR Estrogenic Agents OR Agents, Estrogenic OR Estrogen OR Estrogen Receptor Agonists OR Agonists, Estrogen Receptor OR Receptor Agonists, Estrogen OR Estrogen Effect OR Estrogenic Effect OR Estrogenic Effects OR Effects, Estrogenic OR Estrogen Effects OR Effects, Estrogen OR Estrogens, Conjugated OR Estradiol OR Estradiol 17 beta OR Progestins OR Progestagenic Agent OR Effect, Progestin OR Progestin Effect OR Hormones OR Hormones, Hormone Substitutes, and Hormone Antagonists OR Hormones, Gonadal OR Gonadal Hormone OR Hormone, Gonada OR Gonadal Steroid Hormones OR Estrogen Receptor Modulators OR Steroid Hormones, Gonadal OR Sex Steroid Hormone OR Hormone, Sex Steroid OR Steroid Hormone, Sex OR Sex Steroid Hormones OR Hormones, Sex Steroid OR Steroid Hormones, Sex OR Sex Hormone OR Hormone, Sex OR Sex Hormones OR Gonadal Steroid Hormone OR Estrogen Replacement Therapies OR Replacement Therapies, Estrogen OR Therapies, Estrogen Replacement OR Therapy, Estrogen Replacement OR Replacement Therapy, Estrogen OR Estrogen Replacement OR Estrogen Replacements OR Replacements, Estrogen OR Replacement, Estrogen OR Postmenopausal Hormone Replacement Therapy OR Hormone Replacement Therapy, Post-Menopausal OR Hormone Replacement Therapy, Post Menopausal OR Estrogen Progestin Replacement Therapy OR Estrogen Progestin Combination Therapy OR Raloxifene Hydrochloride OR Therapy, Hormone Replacement OR Hormone Replacement Therapies OR Replacement Therapies, Hormone OR Therapies, Hormone Replacement OR Replacement Therapy, Hormone)) AND (Article) AND(TS=(Pulmonary Neoplasms OR Neoplasms, Lung OR Lung Neoplasm OR Neoplasm, Lung OR Neoplasms, Pulmonary OR Neoplasm, Pulmonary OR Pulmonary Neoplasm OR Lung Cancer OR Cancer, Lung OR Cancers, Lung OR Lung Cancers OR Pulmonary Cancer OR Cancer, Pulmonary OR Cancers, Pulmonary OR Pulmonary Cancers OR Cancer of the Lung OR Cancer of Lung OR Bronchogenic Carcinoma OR Bronchogenic Carcinomas OR Carcinomas, Bronchogenic OR Carcinoma, Bronchial OR Bronchial Carcinoma OR Bronchial Carcinomas OR Carcinomas, Bronchial OR Small Cell Lung Cancer OR Oat Cell Lung Cancer OR Small Cell Cancer Of The Lung OR Carcinoma, Small Cell Lung OR Oat Cell Carcinoma of Lung OR Carcinoma, Non Small Cell Lung OR Carcinomas, Non-Small-Cell Lung OR Lung Carcinoma, Non-Small-Cell OR Lung Carcinomas, Non-Small-Cell OR Non-Small-Cell Lung Carcinomas OR Non-Small-Cell Lung Carcinoma OR Non Small Cell Lung Carcinoma OR Carcinoma, Non-Small Cell Lung OR Non-Small Cell Lung Carcinoma OR Non-Small Cell Lung Cancer OR Non small Cell Lung Cancer))

Embase

('estrogen therapy' OR 'hormonal therapy' OR progesterone OR 'hormone substitution') AND 'lung cancer' AND [1966-2022]/py

Cochrane

#1 MeSH descriptor: [Lung Neoplasms] explode all trees

#2 MeSH descriptor: [Hormone Replacement Therapy] explode all trees
#3 lung cancer
#4 hormone replacement therapy
#5 estrogen replacement therapy
#6 estrogen
#7 progesterone
#8 (#1 OR #3)
#9 (#2 OR #4 OR #5 OR #6 OR #7)
#10 (#8 AND #9)
with Publication Year from 1800 to 2022, with Cochrane Library publication date from Jan 1800 to Dec 2022, in Trials

Scopus
(TITLE-ABS-KEY ("Gonadal Hormones") OR TITLE-ABS-KEY ("Gonadal Steroid Hormones") OR TITLE-ABS-KEY ("Estrogen Receptor Modulators") OR TITLE-ABS-KEY ("Estrogen Replacement Therapy") OR TITLE-ABS-KEY ("Hormone Replacement Therapy")) AND TITLE-ABS-KEY ("lung cancer"))

China National Knowledge Infrastructure
(主题=肺癌) OR (关键词%肺癌) OR (篇关摘%肺癌) AND (篇关摘=激素替代治疗)

WANGFANG
全部:(肺癌) and 全部:(激素替代)) and Date:*-2022

VIP
任意字段=肺癌 AND 任意字段=激素替代治疗 AND 年份：收录起始年-2022

https://dx.doi.org/10.21037/jtd-22-48

Appendix 2

Inclusion criteria (PICO format)

Population	The cohort study included women of any age without lung cancer before the trial; The case-control study included female participants of any age Participants in cases (participants with lung cancer) Participants in controls (participants without lung cancer); Non-smokers were defined as someone who had never smoked or smoked less than 6 months or less than 100 cigarettes during his/her lifetime;
Intervention	Different HRT use (ever/current/former HRT use) Hormone Replacement Therapy (HRT) is a general term used to describe exogenous estrogen therapy, estrogen plus progesterone therapy or other hormone therapy. Current HRT users were defined as those who used HRT after the baseline (the baseline was defined as the time at which the participants entered into the cohort studies). Former users were defined as those who had used HRT at any time before the baseline. Ever users were defined as those who used HRT but no indication of when they used it, and included former users and current users
Comparator	No HRT use
Outcomes	The incidence of lung cancer; The selected articles should report risk estimates [odds ratio (OR), relative risk (RR), or hazard ratio (HR)] and 95% confidence intervals (CIs) or provide information available for us to calculate relevant risk estimates